BA Portfolio Project

Retail Stock-Out Reduction

A Business Analyst case study using a real retail inventory dataset to identify stock-out risks, forecast inaccuracies, promotion inefficiencies, and pricing gaps across 5 stores and 20 products.

73,100	5	20	5
Dataset Rows	Stores	Products	Deliverables

Project	Retail Stock-Out Reduction
Analyst	BanothDeepak
Dataset	Retail Store Inventory Forecasting (Kaggle)
Tools	Python · pandas · matplotlib · VS Code · GitHub
Phases	Requirements → Data Analysis → Dashboard → Portfolio
Date	May 2026

1. Business Problem

A retail chain operating 5 stores across multiple regions is experiencing simultaneous stock-out and overstock situations. The initial assumption was that this was a category-specific issue — but analysis reveals a systemic replenishment model failure affecting all 5 categories equally.

> **Key Finding:** Every category sits at ~33% low-stock AND ~37% overstock simultaneously — a replenishment model failure, not a product problem.

2. Analytical Approach

Phase	Name	Output
Phase 1	Project Scoping	Defined project title, scope, and business context for the retail inventory domain.
Phase 2	Requirements Gathering	Built a Data Dictionary (15 columns), 5 User Stories with acceptance criteria, and a
Phase 3	Data Analysis	Python scripts (pandas + matplotlib) covering all 5 user stories. Outputs: CSVs, PN
Phase 4	Dashboard & Deck	8-slide PowerPoint deck + interactive HTML dashboard with 7 tabs and native char
Phase 5	Portfolio Packaging	PDF case study, LinkedIn post, Notion template, and GitHub push.

3. User Stories & Key Findings

ID	Persona	Business Need	Key Finding
US-01	Store Ops Manager	Flag products where Inventory < Units Sold x 1.5	~4,800 low-stock rows per category — systemic, not isolated
US-02	Supply Chain Planner	Compare forecast vs actual by season & category	MAE flat at 8.32-8.40 — zero seasonal weighting in model
US-03	Category Manager	Compare inventory & sales on promo vs normal days	137.8 vs 137.6 units — promotions deliver no uplift
US-04	Pricing Analyst	Price gap vs competitor by region & category	Groceries at -18.3% — exceeds 15% underpriced threshold
US-05	Operations Director	% overstock & low-stock per store & region	Consistent 38% overstock across all 5 stores

4. Data Quality Observations

A key BA skill is flagging data anomalies during analysis. Two issues were identified and documented:

DQ-01 Negative Demand Forecast values: The Demand Forecast column contains negative values (min: -10.07). These were excluded from US-02 forecast accuracy analysis and flagged in the BRD as a data governance issue requiring upstream investigation.

DQ-02 No Reorder Point column: The dataset contains no explicit Reorder Point field. A derived threshold was calculated as Units Sold x 1.5 per US-01 acceptance criteria, documented in the Data Dictionary as a calculated field.

5. Recommendations

01

Fix the Replenishment Model

Define dynamic reorder points using Inventory Level vs Units Sold x 1.5. The simultaneous 33% low-stock and 37% overstock across all categories confirms the formula is broken — not the products or stores.

02

Introduce Seasonal Forecast Weighting

MAE of 8.32-8.40 is flat across all seasons — no seasonal adjustment exists. Prioritise Furniture (Autumn) and Toys (Winter) for model recalibration first.

03

Conduct Promotion ROI Audit

Promotions run on 50% of all trading days yet deliver zero incremental demand (137.6 vs 137.8 avg units sold). Audit all promotion types before next cycle.

04

Review Grocery Pricing Nationally

Groceries average -18.3% below competitor price — exceeding the 15% alert threshold across all regions. A national pricing policy review is required to recover margin.

6. Portfolio Deliverables

#	Artefact	Format	Phase
1	Data Dictionary — 15 columns defined with business language	Doc	2
2	5 User Stories — with personas, outcomes & acceptance criteria	Doc	2
3	Business Requirements Document — 9 sections	DOCX	2
4	Phase 3 Python Analysis — all 5 user stories	PY + CSV + PNG + XLSX	3
5	Phase 4 PowerPoint Deck — 8 slides	PPTX	4
6	Phase 4 Interactive Dashboard — 7 tabs	HTML	4
7	Phase 5 PDF Case Study	PDF	5
8	GitHub Repository — full project history	github.com	All

GitHub Repository: github.com/BanothDeepak/retail-stockout-ba-project